Imaging Data: Adding a new type of annotations to 3DBionotes-WS

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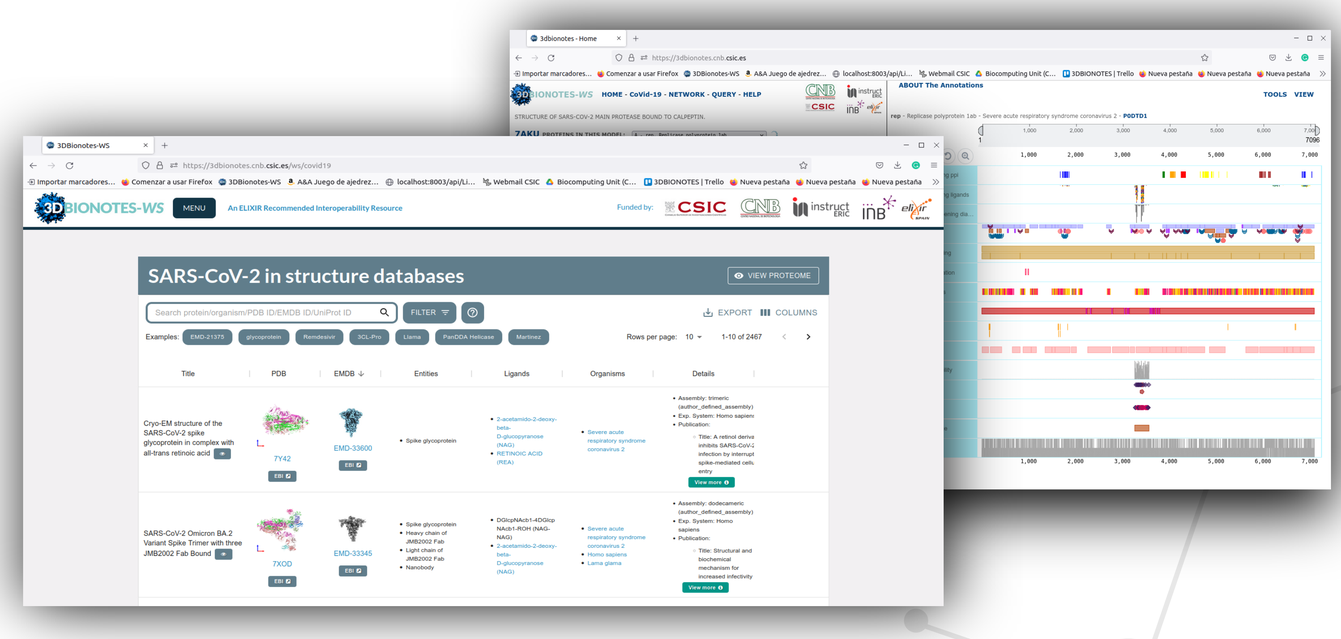
3DBionotes-WS (Macias et al., 2021) is a set of Web Services that provides multiple annotations oriented to structural biology analysis plus a website interface that includes a 3D viewer for macromolecular models and Cryo-EM derived maps.

Current annotations include domain families, genomic variations associated with diseases, immune epitope sites, short linear motifs, disordered regions and post-translational modifications that can be explored interactively at the sequence and structural level.

After the COVID-19 outbreak, easy access to SARS-CoV-2 data and other related viruses became key to enabling the early development of vaccines and potential antiviral drugs. Aiming to collect and provide all available structural information, a new section was set up on 3DBionotes-WS: the COVID-19 Structural Hub.

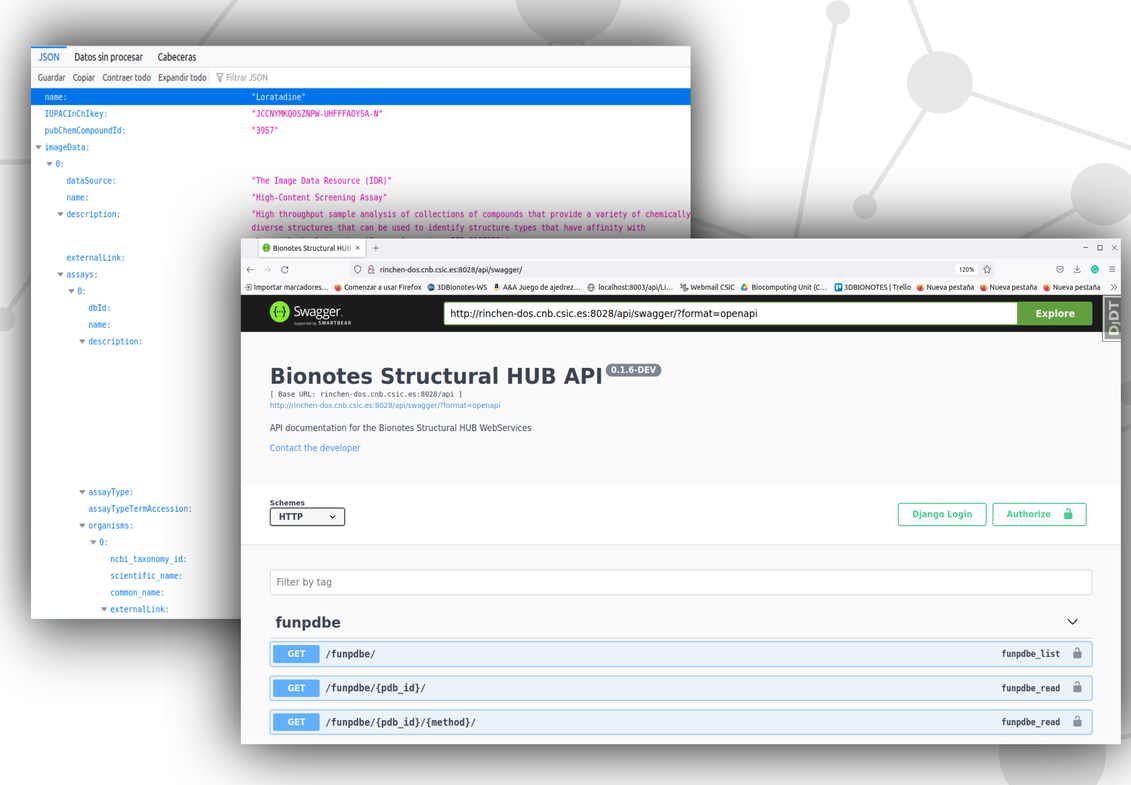
Atomic models from the Protein Data Bank (PDB) and volume maps from the Electron Microscopy Database (EMDB) are displayed here together with key information about interactions with other proteins or with small molecules identified in drug screening studies.



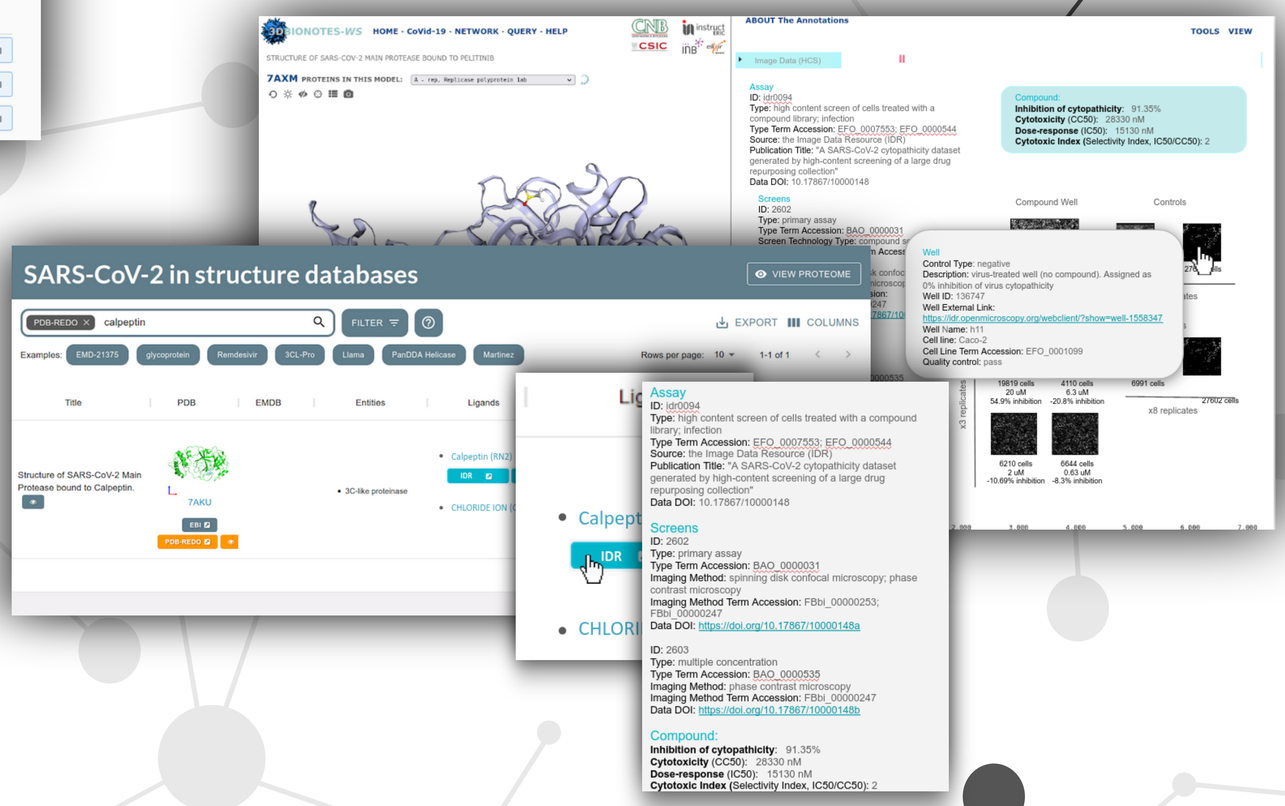
The imaging data field has grown rapidly in content and complexity and consequently, it has become crucial to create an infrastructure for image data standardization, deposition, and sharing. Among the initiatives that have arisen to fulfill this need, there are the Image Data Resource (IDR) and the BioImage Archive (BIA).

As a new set of annotations, our recent efforts have been focused on resulting images from High-Content Screening (HCS) experiments for SARS-CoV-2 deposited at these repositories. 3DBionotes-WS users will be able to search for COVID-19-related macromolecular structures bound to a collection of compounds with potential clinical use.

Third-party tools will also be able to access this data through the new Application Programming Interface (API), whose documentation is provided using the Swagger UI tool.



Currently, imaging data from HCS experiments are available through the 3DBionotes-WS API. As a future enhancement, we are working on the design and implementation of new viewers for HCS imaging annotations both within the 3DBionotes-WS and the COVID-19 Structural Hub.



Jose Ramon Macias, Ruben Sanchez-Garcia, Pablo Conesa, Erney Ramirez-Aportela, Marta Martinez Gonzalez, Carlos Wert-Carvajal, Alberto M Parra-Perez, Joan Segura Mora, Sam Horrell, Andrea Thorn, Carlos O S Sorzano, Jose Maria Carazo, 3DBionotes COVID-19 edition, Bioinformatics, Volume 37, Issue 22, 15 November 2021, Pages 4258–4260. https://doi.org/10.1093/bioinformatics/btab397

https://3dbionotes.cnb.csic.es/ws/api/

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